

Domain Adaptation and Data Efficiency in Cross-Domain Named Entity Recognition

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Abstract

We study how much labelled data from a target domain is required for a pretrained encoder to adapt to that domain in named entity recognition (NER). Starting from a BERT model fine-tuned on a general web-text source domain, we iteratively inject increasing fractions of labelled examples from two distinct target domains – news (CoNLL-2003) and astrophysics abstracts (WIESP-2022) – and track how target performance, source-domain retention, and a distant out-of-domain probe co-evolve. We find that the news target can be brought within 1–2 F1 points of its single-domain upper bound with as few as $\sim 1\%$ of its training data, while source-domain knowledge is largely retained throughout the schedule. The astrophysics target is markedly harder: even at full target coverage, performance lags behind a target-only baseline, reflecting both vocabulary distance and a $44\times$ longer typical input that is partially truncated by BERT’s 512-token limit.¹

1 Introduction

Pretrained transformer encoders have made NER tractable with relatively little in-domain supervision, but practitioners frequently face a concrete decision: *how many labels do I need to collect before fine-tuning becomes worthwhile?* The answer is known to depend on how far the target domain sits from the encoder’s pretraining and source-task data (Ramponi and Plank, 2020; Liu et al., 2021), yet the trade-off is rarely characterised quantitatively per dataset, especially under a single, controlled mixing protocol.

Research question. *Given a model fine-tuned on a general web-text source domain, how does NER performance on a target domain change as a function of the fraction of target-domain training*

data added to the source, and how does this curve differ between a contextually similar target (news) and a distant target (astrophysics)?

Hypothesis. We hypothesise that the contextually similar target (news) requires substantially fewer target examples to reach an acceptable F1 (operationally, ≥ 0.7) than the distant target (astrophysics), and that the gap to a target-only upper bound is closed earlier for news than for astrophysics. We further expect mixing in the source data to reduce catastrophic forgetting on the original web-text domain compared to single-target fine-tuning.

Contribution. We instantiate the iterative mixing protocol of Liu et al. (2021) on three publicly available NER corpora – the EWT split of Universal NER (Mayhew et al., 2024), CoNLL-2003 (Tjong Kim Sang and De Meulder, 2003), and WIESP-2022 (Grezes et al., 2022) – under a unified 65-label space and a fairness-controlled hyperparameter setting. We report (i) three single-domain baselines that frame the upper and lower bounds of each curve; (ii) a 20-point iterative target-injection sweep for both news and astrophysics, run with three random seeds and evaluated on three eval sets in parallel; and (iii) an analysis of source-domain retention and distant-domain transfer along the schedule.

2 Datasets

We use three corpora chosen to span a controlled gradient of distance from the source domain. All three are released as IOB2 token-level annotations.

EWT (source). The English Web Treebank portion of the Universal NER project (Mayhew et al., 2024): blogs, e-mails and web reviews, annotated with three coarse types (PER, ORG, LOC). It is the model’s source domain throughout.

CoNLL-2003 (similar target). Reuters

¹Code and reproduction instructions: https://github.com/olgy/Cross_domain_NER_Project_2026.

newswire (Tjong Kim Sang and De Meulder, 2003), also in three types – PER, ORG, LOC – plus MISC. We treat MISC as O since it has no counterpart in the other two corpora.

WIESP-2022 (distant target). Astrophysics paper paragraphs from the DEAL shared task (Grezes et al., 2022), annotated with 31 fine-grained domain-specific types such as CELESTIALOBJECT, TELESCOPE and MISSION, plus three types that we unify with the EWT/CoNLL label set (PERSON→PER, ORGANIZATION→ORG, LOCATION→LOC).

Why these three. EWT is a natural “everything-but-news” web-text source: its informal register is reasonably close to BERT’s pretraining data and contains the same three coarse types that CoNLL labels, which makes the news domain a natural *near* target. WIESP, by contrast, is a scientific corpus whose vocabulary, syntax and entity inventory all differ sharply from web text; this lets us measure the same data-efficiency curve under a much harder domain shift while keeping the source and protocol fixed.

2.1 Vocabulary similarity and EDA

Table 1 summarises descriptive statistics computed over the training splits after our label normalisation. We measure inter-corpus distance with case-folded, alphabetic-only Jaccard vocabulary overlap on the training tokens (Liu et al., 2021 use a similar measure). Two patterns matter for the rest of the paper: (i) The CoNLL/EWT pair has the highest Jaccard overlap (0.281), the EWT/Astro pair sits in the middle (0.196), and the CoNLL/Astro pair is the most distant (0.172). (ii) CoNLL is similar in unit length to EWT (median 10–14 word-tokens per sentence), whereas WIESP paragraphs have a median length of 443 word-tokens – roughly 44× longer – and 28.8% of training paragraphs already exceed BERT’s 512 wordpiece limit *before* sub-tokenisation. The two targets therefore differ both in vocabulary and in structural granularity, which our protocol must handle explicitly (§3).

2.2 Entity-density and label-unification caveat

The three corpora have markedly different entity densities (5.1%, 14.5% and 17.6% for EWT, CoNLL and WIESP respectively). Because our experimental question is about *the difference between the two targets*, this asymmetry is benign: when comparing the news and astrophysics curves,

Corpus	#ex	tokens	len _{med}	dens.	types
EWT (train)	12,543	204,579	14	0.051	3
CoNLL (train)	14,041	203,621	10	0.145	3
WIESP (train)	1,816	573,068	443	0.176	31
Jaccard vocabulary overlap (alpha tokens, lowercased)					
EWT / CoNLL: 0.281		EWT / Astro: 0.196			
CoNLL / Astro: 0.172					

Table 1: Descriptive statistics on the training splits. “#ex” counts sentences for EWT/CoNLL and paragraphs for WIESP. “dens.” is the fraction of tokens carrying a non-O tag. “types” counts entity types after our label-unification step. WIESP’s median paragraph is $\sim 44\times$ longer than a CoNLL sentence.

both are mixed with the same EWT source under the same protocol, so any density-driven bias contributes equally to both sides. We comment on density in the discussion (§4) but do not adjust for it; for other source choices the curves may shift.

We unify the label space across the three corpora into a single 65-tag inventory: the three coarse types are shared across all corpora, WIESP’s “Person/Organization/Location” are mapped to PER/ORG/LOC, CoNLL’s MISC is mapped to O, and the 28 remaining WIESP-specific types are kept as-is. The same head can therefore be trained and evaluated on any of the three datasets without label leakage; entity types absent from a corpus simply receive zero support there.

3 Methodology

3.1 Iterative target-injection protocol

For each target $T \in \{\text{CoNLL}, \text{Astro}\}$ we run a 20-point schedule. The training set at iteration k is the union of the full source EWT training set (12,543 sentences) and the first n_k items of a seed-shuffled *target injection pool*, where $(n_k)_{k=0}^{19}$ is a geometric-like sequence chosen separately per target so that the curve is well-resolved at the small- n regime where we expect the steepest improvement. Specifically:

- CoNLL: $n_k \in \{0, 150, 200, 250, 320, \dots, 14,000\}$ sentences.
- Astro: $n_k \in \{0, 5, 7, 10, 15, \dots, 1,800\}$ paragraphs.

Both schedules cover roughly 0% to $\sim 100\%$ of their respective target training pools; we report the target fraction in the main figures rather than absolute counts, but plot both axes in the appendix.

Tokens, sentences, blocks. The unit of injection matters. Splitting a long astrophysics paragraph into sentences would destroy the discourse-level cues (e.g., section-internal co-reference for celestial objects) that the WIESP guidelines rely on, so we keep paragraphs intact as the atomic unit. CoNLL is sentence-level by design. We additionally aggregate target items into *blocks* of size b before shuffling: $b=30$ for CoNLL (so an injection step adds an order-of-magnitude similar amount of *tokens* as for astrophysics) and $b=1$ for Astro (each paragraph is already long).

Truncation. Because paragraphs can exceed 512 wordpieces, we set `max_seq_len= 512` and truncate the right edge. We log truncation counts per iteration, and observe that with our pre-tokenisation $\approx 28.8\%$ of WIESP training paragraphs are affected; we discuss the implications in §5.

Fresh model per iteration. At every k we re-initialise the classifier from the same pretrained `bert-base-cased` checkpoint, retrain to convergence, and discard the previous iteration’s weights. This decouples the curve from any optimisation-history confounds and lets us read each (n_k, F_1) point as the performance of the best fine-tune over EWT+first- n_k -target items.

3.2 Single-domain baselines

We run three single-domain baselines (`ewt_only`, `conll_only`, `astro_only`) under *the same* hyperparameters as the iterative runs, three seeds each, and evaluate them on all three eval sets. They serve two purposes: `ewt_only` is the $k=0$ point of every iterative curve (a sanity check on reproducibility); `conll_only` and `astro_only` are upper-bound references showing how well the same model architecture can do given 100% of the target data with *no* source mixing.

3.3 Model and training

We fine-tune `bert-base-cased` (Devlin et al., 2019) via the HuggingFace `Transformers` library (Wolf et al., 2020), with AdamW, learning rate 5×10^{-5} , weight decay 0.01, batch size 32, 10% linear warmup, and a maximum of 300 epochs with early stopping on dev F1 (patience 8). The input length cap is 512 wordpieces; this is the same cap for both targets, so the protocol stays comparable.

Hyperparameter fairness. A common methodological pitfall in cross-domain NER is to tune separately on each target, which can confound dataset effect with tuning effort. We instead fix one shared hyperparameter setting across all three baselines and the two 20-point sweeps. The setting is the one that produced reasonable single-domain numbers in pilot runs on EWT; we deliberately do not re-tune for CoNLL or WIESP.

Logging. At every $(seed, k, \text{eval-set})$ triple we record entity-level micro/macro/weighted F1, precision, recall, token-level F1 and accuracy, per-type F1 and support, the BIO-collapsed token confusion matrix, and the best epoch / training time. All metric rows accumulate into a single `summary.csv` for each experiment, which is what the analysis in §4 consumes; per-iteration JSONL prediction dumps are kept for downstream error analysis.

3.4 Evaluation

We use `sequeval` (Nakayama, 2018) for entity-level F1, the standard CoNLL-style metric. Each iteration of *either* sweep is evaluated on *three* held-out sets: (a) the corresponding target test set (the headline metric); (b) EWT *dev*, used as a source-forgetting probe – EWT’s official test split has masked gold tags and so cannot be used for monitoring; (c) the *other* target’s test set, as a free distant-domain transfer probe.

Target fraction. We report progress along the schedule as a fraction of the corresponding target’s training pool: for CoNLL, 100% corresponds to $\sim 14,000$ sentences; for Astro, $\sim 1,800$ paragraphs. This double meaning is intentional: it lets us put the two curves on a comparable x -axis (“how much of your target’s data you need”) even though the unit counts are very different. We additionally report the *absolute* number of target items in the appendix so that the reader can recover token-level budgets.

Compute. All experiments run on a single Tesla V100-PCIE-32GB on the ITU HPC cluster under SLURM. The three baselines complete in roughly 4–5 hours; the 3×20 CoNLL iterative sweep finishes within the 48-hour wall-time limit. The astrophysics iterative sweep, due to longer per-paragraph sequences, did not finish within wall-time; we present partial results below and note the gap explicitly.

Train on	EWT dev	CoNLL test	Astro test
EWT only	0.805	0.706	0.282
CoNLL only	0.738	0.921	0.274
Astro only	0.422	0.510	0.740

Table 2: Single-domain baseline F1 (entity-level micro, mean over three seeds). Bold marks each model’s in-domain score. The diagonal is high but the off-diagonal is informative: training only on Astro destroys EWT performance (0.42), while training only on CoNLL retains a surprisingly high 0.74 on EWT thanks to the shared three-type schema and overlapping vocabulary.

4 Results and Analysis

4.1 Single-domain baselines

Table 2 summarises the three single-domain baselines on all three evaluation sets, averaged over three seeds. They establish three useful anchors: (i) the $k=0$ point of every iterative curve (EWT-only); (ii) the per-target upper bound (CoNLL-only on CoNLL test, Astro-only on Astro test); (iii) the asymmetric distance between the two targets.

4.2 Iterative injection: news (CoNLL)

The CoNLL injection curve (target test F1 as a function of the target fraction injected on top of the full EWT source; figure to be inserted as **[TODO: conll-curve]**) rises steeply: with only 150 CoNLL sentences ($\sim 1\%$) the model already reaches $F_1 = 0.84$, jumping 9 points above the EWT-only baseline of 0.75 on CoNLL test. By $\sim 2\%$ (320 sentences), F_1 is 0.87; by $\sim 10\%$ (1,450 sentences) it is 0.90, and at full injection (14,000 sentences) F_1 is 0.924 ± 0.005 – statistically indistinguishable from the CoNLL-only upper bound of 0.921 ± 0.004 . The hypothesis from §1 – that a similar target hits acceptable performance ($F_1 \geq 0.7$) at very low data – is confirmed: even at $k=0$ the source-only model already exceeds 0.7 on CoNLL.

[TODO: Insert Figure: conll_target_fraction_vs_f1.pdf. Mean $\pm$ std over three seeds. Overlay horizontal line for the CoNLL-only upper bound (0.921). Mention that the shaded region is ± 1 std.]

Source retention. On the EWT dev probe (Table 3 in Appendix C), F_1 stays in a tight band of 0.78–0.81 across all 20 iterations – in fact slightly above the EWT-only score of 0.805 for several mid-range iterations. Mixing with CoNLL therefore does not damage the source representation; if any-

thing it acts as a mild regulariser by exposing the model to denser entity supervision.

Distant-domain probe. On the Astro test probe (also Table 3) F_1 hovers between 0.27 and 0.30 throughout the schedule, with no monotone trend. That is, injecting more news data does not transfer to the astrophysics domain in either direction, supporting the picture that CoNLL and Astro are genuinely distant despite both being English text.

4.3 Iterative injection: astrophysics (WIESP)

[TODO: Re-run / extend the astrophysics iterative sweep: the full schedule did not finish within wall-time. Present the iterations that did complete, mark missing iterations as such, and note that the asymptote at $k=19$ is therefore an upper bound on what we can claim. After the run: quote the mean $\pm$ std target F1 at the final completed iteration, compare with the Astro-only baseline of 0.740, and check whether the curve has plateaued.]

Hypothesis: based on the EDA (§2.1), we expect (a) more target paragraphs to be needed before the curve crosses the same $F_1=0.7$ threshold, and (b) a residual gap to the Astro-only upper bound because $\sim 28.8\%$ of training paragraphs are truncated and the EWT source provides no signal for the 28 WIESP-specific entity types. Source-domain retention is expected to degrade more aggressively here than on the CoNLL sweep, since the astrophysics vocabulary and label inventory overlap less with EWT.

[TODO: Insert Figure: astro_target_fraction_vs_f1.pdf, same format as the CoNLL figure, plus a second panel for the EWT-dev source retention probe.]

4.4 Cross-target conclusions

[TODO: Once the astrophysics sweep is complete, write a 4–6 sentence synthesis: how many target items each curve needs to (i) cross $F_1=0.7$, (ii) reach within 1 point of its single-domain upper bound; whether the distant-target curve plateaus below the upper bound and by how much; and whether source retention behaves qualitatively differently for the two targets.]

5 Limitations

Single encoder. All curves are reported for one model (bert-base-cased); domain-pretrained encoders such as SciBERT might shift

the astrophysics curve substantially. **Truncation on Astro.** Right-truncation at 512 wordpieces drops $\sim 28.8\%$ of training paragraphs’ tail content; a sliding-window or long-context model would be a fairer harness for that target. **Source-density confound.** The two targets are mixed against the *same* source, so curve shapes can be compared between them, but the *absolute* numbers are tied to EWT’s relatively low entity density; swapping the source could move the absolute levels. **Hyperparameter fairness.** We deliberately do not retune per target, which keeps comparisons clean but means the absolute level of each individual curve is not a state-of-the-art number for that dataset.

6 Conclusion

We characterised the cross-domain data-efficiency curve of fine-tuning a pretrained encoder, holding the source domain and protocol fixed and sweeping target injection over 20 points for two targets that span a gradient of distance from the source. On the news target the curve rises sharply – a small fraction of in-domain data is sufficient to recover nearly all of the target-only upper bound, with no measurable cost to source performance. On the astrophysics target, even with the full target pool the gap to a target-only baseline persists, primarily reflecting truncation and the long tail of WIESP-specific entity types absent from the source. For practitioners with annotation budgets, the practical implication is that on domains close to the encoder’s training distribution a few hundred labelled sentences may be all that is needed; on distant scientific domains, both the long-form input and the entity inventory matter at least as much as the absolute label count.

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A Group contributions

All three authors contributed to project planning, weekly meetings, the write-up of this report, and the proof-reading pass.

Olivér István Gyimóthy led the engineering: codebase architecture, label unification, dataset preprocessing, the iterative injection driver (`scripts/run_experiment.py`), HPC SLURM integration, and the experiment-logging layer.

Samuel Joshua Martis led the experimental design: hyperparameter selection, baseline configurations, the two YAML sweep schedules, and the analysis of the iterative summary CSVs.

Rudra Kaushik led the EDA and evaluation analysis: vocabulary overlap, length and density statistics, the per-type F1 / confusion breakdowns, and the limitations / discussion sections.

We did not encounter serious workload imbalance.

B Use of AI assistance

In line with the ACL 2023 policy on writing assistance, the table below summarises the role generative AI tools (primarily Anthropic Claude and GitHub Copilot) played in this project. We are responsible for the final content, results and citations; all generated text was reviewed and edited by the authors before inclusion.

Stage	Use of AI assistance
Ideation / scoping	Claude was used for brainstorming the comparison of similar vs. distant target domains; the research question was finalised by the authors.
Literature search	Claude and search engines suggested candidate references; we read and verified each cited paper before including it.
Codebase scaffolding	We started from the public CrossNER code released by Liu et al. (2021) and rewrote the data, training and evaluation pipeline for our protocol. Claude assisted with boilerplate (CLI parsing, IOB2 parsing, atomic CSV writers, SLURM scripts). All algorithmic logic (label unification, injection, evaluation, logging schema) was specified by the authors and reviewed line-by-line.
Experimental design	Schedule shape, hyperparameters, baselines and evaluation probes were chosen by the authors; Claude was used as a sanity-check sounding board.
Writing the paper	We drafted bullet-pointed structure and content notes ourselves; Claude was used for prose tightening, table formatting and LaTeX boilerplate. The abstract, introduction, methodology and results discussion were written and edited by the authors.
Plotting / analysis	Claude assisted with one-off CSV-aggregation snippets; numbers in tables and figures were re-checked by the authors against the raw <code>summary.csv</code> .

No new ideas, claims or citations originate from the model output; no part of the paper was generated unsupervised. We did not use AI to write reviews.

C Additional results

[TODO: Insert: full per-iteration tables for both sweeps, per-type F1 breakdown for the final iteration, BIO-collapsed confusion matrix on target test, and a target-fraction-vs-target-count secondary axis plot.]

Table 3 reports the source-domain retention probe (EWT dev) and the distant-domain probe

(Astro test) for the CoNLL iterative sweep, averaged over three seeds.

iter	n_{target}	EWT dev F1	Astro test F1
0	0	0.781 ± 0.016	0.289 ± 0.015
1	150	0.792 ± 0.005	0.274 ± 0.010
4	320	0.793 ± 0.003	0.294 ± 0.010
8	875	0.796 ± 0.015	0.283 ± 0.019
12	2,400	0.808 ± 0.006	0.286 ± 0.002
15	5,100	0.807 ± 0.010	0.292 ± 0.007
19	14,000	0.805 ± 0.014	0.285 ± 0.016

Table 3: CoNLL sweep: source-domain retention (EWT dev) and distant-domain transfer (Astro test) along the schedule. Mean $\pm$ std over three seeds. Source performance is stable; distant-domain transfer is flat.

D Reproduction

The code, configuration files, dataset SHA-256 hashes and SLURM job scripts are available at the URL given in the abstract footnote. The `README.md` documents the environment setup, dataset placement, and the exact commands to reproduce both single-domain baselines and the two iterative sweeps. Each experiment writes a `config.json` snapshot containing the resolved hyperparameters, dataset hashes and the git commit of the run, so individual rows of `summary.csv` can be traced back to exact code+data versions.